Lab 5

Name: **JISHNU SIVAKUMAR** SRN: **PES2UG24CS204**

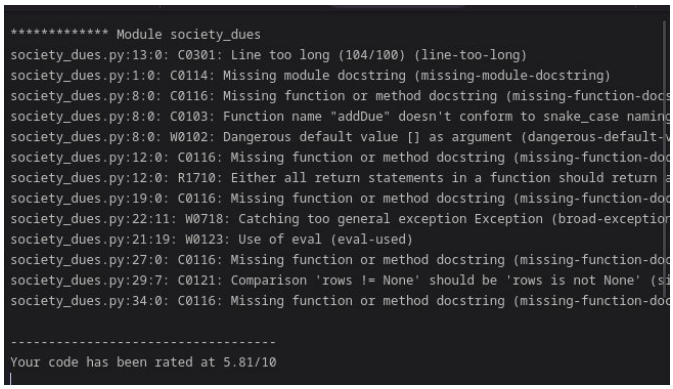
ISSUES

Issue	Type	Line(s)	Description	Fix
Hard-coded password	Security	5	Password stored in source.	Removed the unused constant.
SQL query construction	Security	9	Values inserted into SQL using an f-string.	Used SQL placeholders and parameters.
Mutable default list	Bug	8	A default list could be shared between calls.	Used None and created a list inside the function.
Function name	Style	8	addDue did not follow snake_case.	Renamed it add_due and updated calls.
Missing return	Bug	12–16	collect() had no explicit return for a missing record.	Added return None.
Broad exception	Reliability	22	Catching Exception could hide unrelated errors.	Caught specific exception types.
None comparison	Style	29	Compared rows with None using !=.	Changed it to is not None.
Spacing / long lines	Style	4, 12–15	Spacing and line length warnings.	Improved spacing and split some lines; two E501 warnings remain.

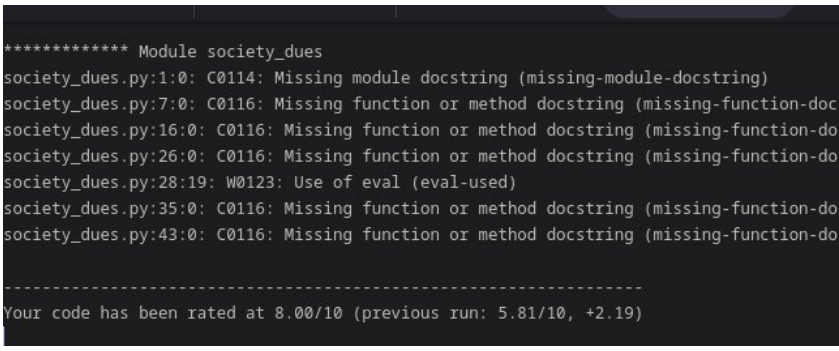
REPORTS before and after fix:

PYLINT:

beofre



after:



FLAKE8:

```
society_dues.py:4:80: E501 line too long (87 > 79 characters)
society_dues.py:12:1: E302 expected 2 blank lines, found 0
society_dues.py:13:80: E501 line too long (104 > 79 characters)
society_dues.py:15:80: E501 line too long (92 > 79 characters)
society_dues.py:29:13: E711 comparison to None should be 'if cond is not None:'
```

```
society_dues.py:4:80: E501 line too long (87 > 79 characters)
society_dues.py:8:80: E501 line too long (81 > 79 characters)
```

BANDIT:

```
Run started:2026-10-14 04:09:19.843209+00:00

Test results:
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'flat#101'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: ./society_dues.py:5:21
4       conn.execute("CREATE TABLE dues (flat TEXT, month TEXT, amount INTEGER, paid INTEGER)")
5       TREASURER_PASSWORD = "flat#101"
6

--------------------------------------------------
>> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query constructio
   Severity: Medium   Confidence: Medium
   CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
   Location: ./society_dues.py:9:19
8       def addDue(flat, month, amount, history=[]):
9           conn.execute(f"INSERT INTO dues VALUES ('{flat}', '{month}', {amount}, 0)")
10          history.append((flat, month))

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: ./society_dues.py:21:19
20          try:
```

```
Run started:2026-10-14 05:16:07.602701+00:00

Test results:
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: ./society_dues.py:28:19
27          try:
28              late_fee = eval(rule)
29          #except Exception:

--------------------------------------------------

Code scanned:
        Total lines of code: 35
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
```

```
20          try:
21              late_fee = eval(rule)
22          except Exception:

--------------------------------------------------

Code scanned:
        Total lines of code: 31
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 1
                Medium: 2
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 2
                High: 1
Files skipped (0):
```

```
Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 1
Files skipped (0):
```